AGENTIC GAME QA · OPEN-SOURCE DEMO

Pointing a game-testing agent harness at a real open-source shooter

Freedoom is a complete, open-source game for the Doom engine. We ran it headless through ViZDoom and pointed four agent types at it: a scripted smoke bot, a random-walk baseline, a curiosity agent that returns to its least-visited frontier, and a combat bot. Every finding below was re-run in a fresh process before it was filed.

Game	Freedoom: Phase 2 (32 maps)
Engine	ViZDoom 1.3.2, headless
Hardware	2 CPU cores, cloud sandbox
Game time	12.0 hours simulated
Budget	120,000 tics (57 min) per map per explorer
Run date	16 Oct 2026

32/32
Maps passed smoke: boot, spawn, move, fire, use, invariants

12.0 h
Game time played by agents, at up to 60× real time

3
Tickets filed with repro; 9 more candidates closed in triage

38% vs 31%
Average map coverage, curiosity agent vs. random walk (3 seeds per map)

01 · FINDINGS

What the agents found, after triage

Every candidate was re-run in a fresh game process before it became a ticket. 12 candidates came out of the runs; 9 were closed in triage (8 test artifacts and one bug in our own checks) and are listed below so you can see why.

HIGH · BLOCKS TEST TOOLING   REPRODUCED · MAP01 AND MAP07   QA-001 · Engine · ViZDoom 1.3.2

Player input is ignored after the second save-state load

After loading the same save state a second time, the next movement and turn commands do nothing. The first load works; every later load leaves the player frozen until one idle tic has run. This silently breaks any tool that rewinds the game with save states. On a short MAP01 run our explorer reached 11% of the map with the defect and 21% once we worked around it, and any stuck-spot detector built on loads reports false positives.

`load 1: moved 586 units` `load 2: moved 0 units` `load 3: moved 0 units` `load 4: moved 0 units` `with 1 idle tic: 595 units every load`

Expected	Each load followed by 20 forward steps moves the player the same distance.
Workaround	Run one idle tic after every load (applied in the harness, see doomqa/env.py).
Upstream	Not mentioned in the documented save/load API. The ViZDoom issue tracker has not been searched yet; worth checking before filing.

```
g.save('s.sav')
for i in range(4):
    g.load('s.sav')
    for _ in range(20):
        g.make_action(FORWARD, 4)   # 4 tics per step
# distance moved per load: 586, 0, 0, 0
```

LOW · AFFECTS REPRO DESIGN   REPRODUCED · DETERMINISTIC   QA-002 · Engine · ViZDoom 1.3.2

A loaded save state is not bit-identical to the live game

Saving, continuing 60 steps, then loading and replaying the same 60 steps ends 0.029 map units from where the live run ended. It is small and deterministic, but it means a run that used save states cannot be rebuilt by replaying its inputs from spawn: each load shifts the trajectory slightly and the shifts add up.

`live end (559.987, -16.015)` `after load (559.958, -16.016)` `drift 0.029 units`

Impact	Bug reports can't rely on "replay these inputs from spawn" once save states were used.
Harness change	Every finding now captures the segment's starting save state and the inputs that followed.

INFO · HOW TO FILE REPROS   DEMOS REPLAYED 24/24 EXACTLY   QA-003 · Test design

Same seed and same inputs don't reproduce a match; engine demos do

Three fresh episodes with the same seed and identical inputs ended with different health (90, 82, 92), because combat randomness is not reset per episode. The engine's own demo recordings replayed each run exactly. All 24 combat-bot episodes were recorded as demos and replayed in a fresh process; final kills, items and damage matched in 24 of 24.

`run 1: health 90` `run 2: health 82` `run 3: health 92` `demo replays match: 3/3`

Recommendation	Attach the demo file (or a save + inputs) to every ticket. Treat input logs as notes, not repro.

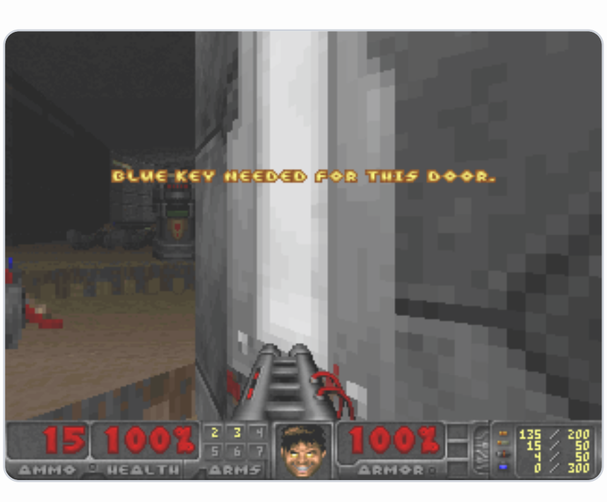
CLOSED · TEST ARTIFACT   REPRODUCED, THEN CLOSED   QA-004 · MAP03 · MAP04 · random agent

8 "player can't move" reports were the god-mode player boxed in by monsters

The random agent stopped moving in a corridor on MAP03 and next to a blue-key door on MAP04. Both replayed exactly from their save states in a fresh process. Triage then found two Demons within about 50 units in each case: in god mode the player can't die and the Demons won't back off, so nobody moves. In normal play the player kills them or dies. The extra seed runs saved six more stuck states; all six had monsters within 72 units, four were still blocked after a 10-second wait, and in the other two the monster had moved off. The stuck detector now lists nearby monsters and files these as "blocked by actors" instead of a level bug.

`MAP03: moved ≤ 1.7 units in 8 headings` `MAP04: moved ≤ 2.1 units in 8 headings` `MAP03: escape test reached 270.6 units` `MAP04: escape test reached 256.3 units`

Monsters nearby	MAP03: Demon at 51 units, Demon at 52 units; MAP04: Demon at 48 units, Demon at 51 units
Replay	MAP03 0.2 units off, still stuck, MAP04 0.0 units off, still stuck
Better setup	For navigation coverage, run explorers with monsters set to ignore the player (notarget) or removed, and keep monsters for the combat bot.
Also found	Our probe's turn command was being dropped (the delta turn button applies a tic late), so the first probe only ever tried one heading. Fixed: turns now use the turn buttons and check the real angle.



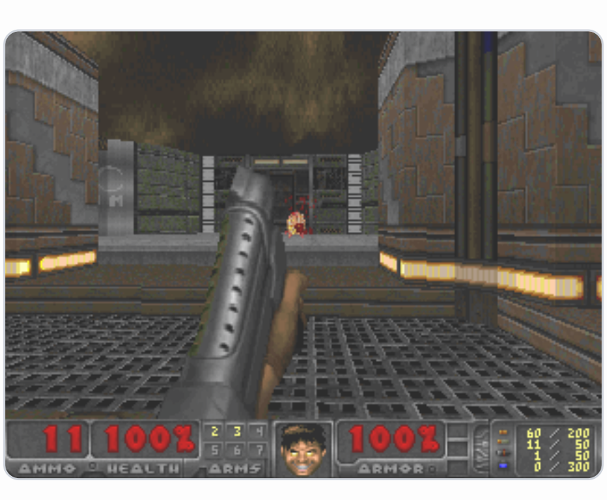
MAP04 at detection: blue-key door ahead, Demons behind

CLOSED · HARNESS BUG   CLOSED DURING TRIAGE   QA-005 · MAP04 · harness

"Player below the floor" on MAP04 was our own lookup error

The floor check reported the player 24 units below the floor of sector 5. An exact geometry test showed the point sits where sector 5's outline overlaps two other sectors, and the player was standing on sector 40, whose floor is exactly at z = −24. The check now uses every sector under the player's footprint, and the finding no longer fires.

Root cause	Per-sector point-in-polygon picked one of several overlapping sectors.
Fix	Compare z with the lowest floor of all sectors under the player's 16-unit radius.



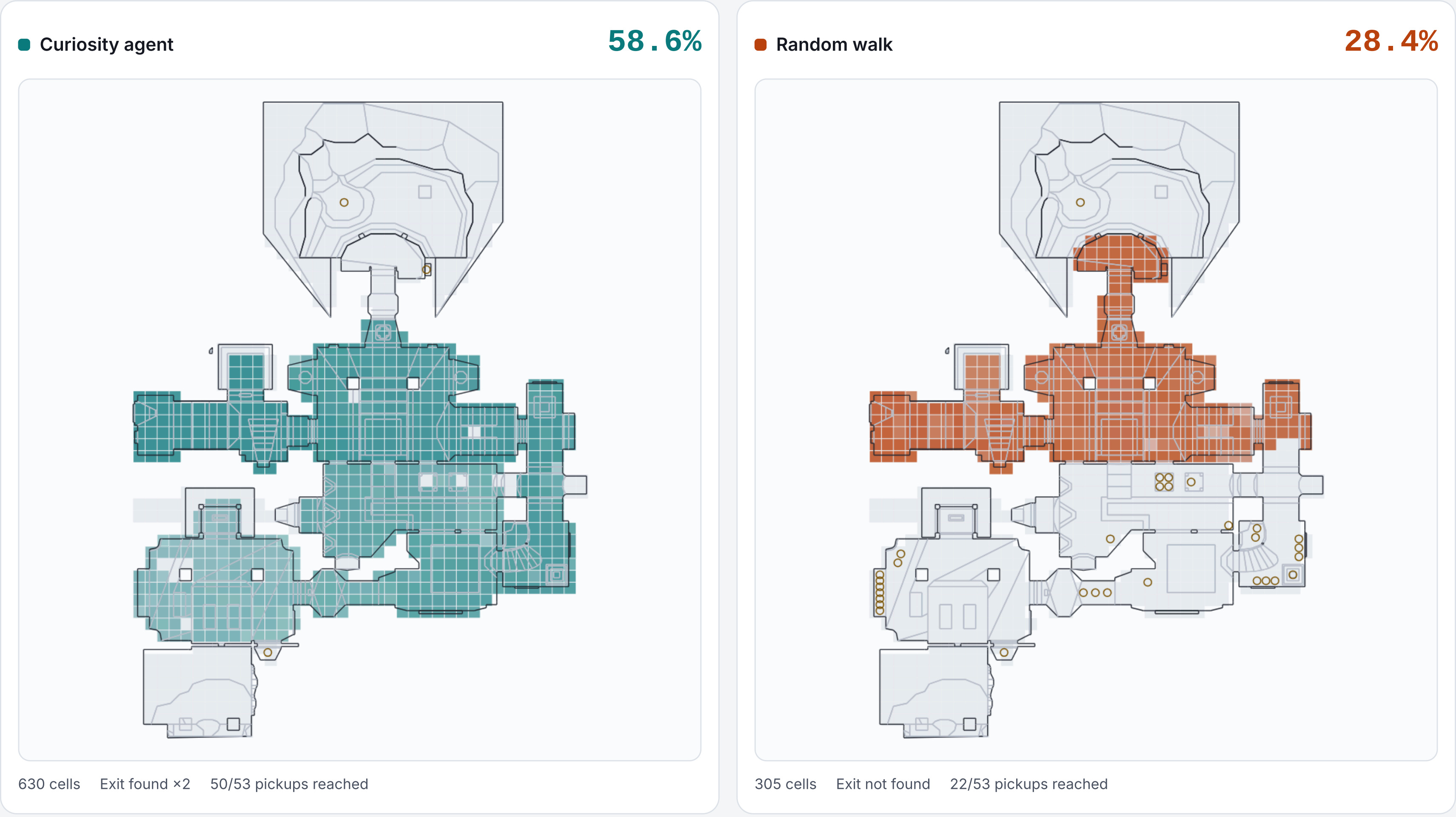
Frame at detection: player on a grate above a lower floor

02 · COVERAGE RACE

Curiosity agent vs. random walk, same budget

The maps below show seed 1; the table averages all seeds. Both agents get the same number of game tics and the same action style. The curiosity agent keeps an archive of save states and goes back to the cells it has visited least (Go-Explore with a count-based novelty weight). Press play or drag the timeline.

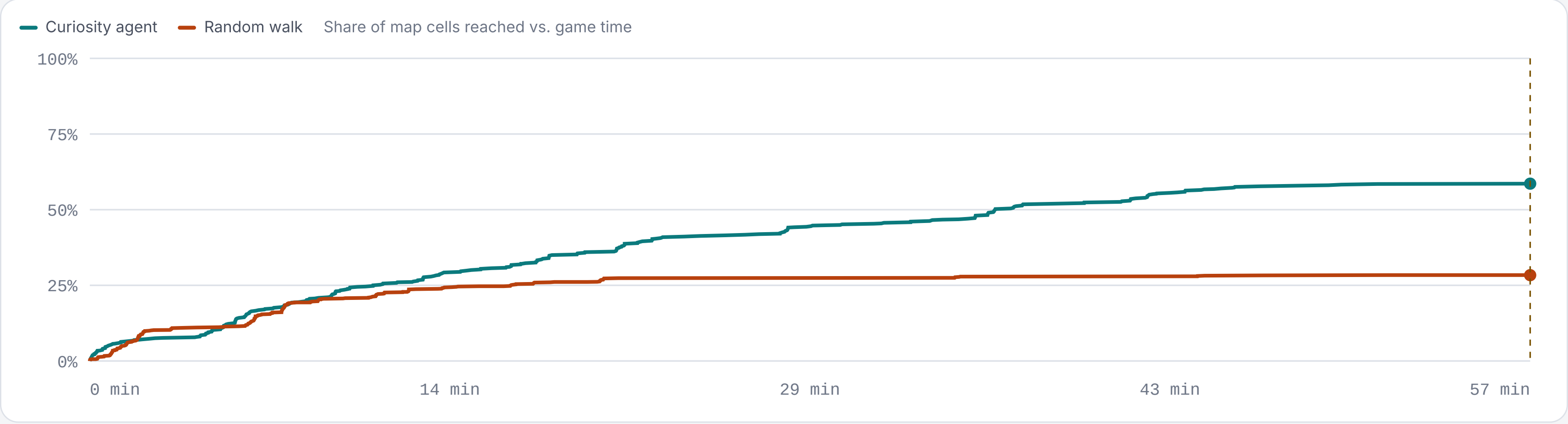
MAP01 MAP02 MAP03 MAP04 MAP05 MAP06



630 cells   Exit found ×2   50/53 pickups reached

305 cells   Exit not found   22/53 pickups reached

`120,000 tics · 57.1 min game time`



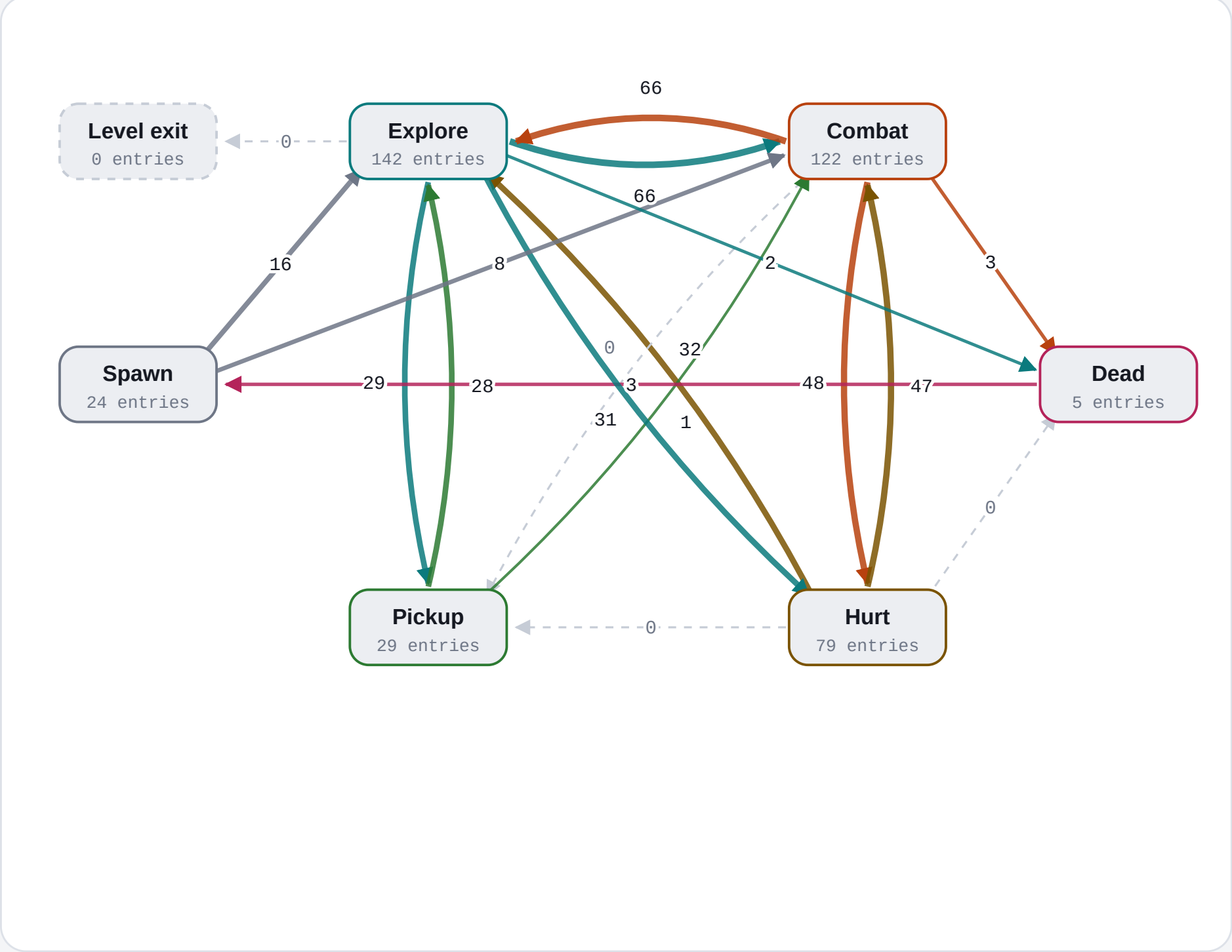
Map	Walkable cells	Curiosity, mean of 3 (range)	Random, mean of 3 (range)	Difference	Runs that found the exit	Pickups reached, seed 1
MAP01	1,012	61.9% (57–70)	35.7% (28–50)	+26.2 pts	1/3 vs 1/3	50/53
MAP02	907	64.4% (61–66)	50.0% (32–59)	+14.4 pts	0/3 vs 0/3	69/86
MAP03	1,912	14.6% (12–20)	19.0% (12–25)	-4.4 pts	0/3 vs 0/3	24/62
MAP04	1,088	38.9% (38–40)	41.1% (41–41)	-2.2 pts	0/3 vs 0/3	52/136
MAP05	1,400	28.1% (21–42)	22.9% (19–30)	+5.1 pts	0/3 vs 0/3	42/103
MAP06	1,149	19.2% (19–19)	19.3% (19–19)	-0.1 pts	0/3 vs 0/3	4/56

MAP06 hits the same ceiling in every run for both agents (19.2–19.3%): the area reachable from spawn stays closed until something is triggered that neither agent finds. That is the cue for a goal-directed persona (keys, switches) rather than more exploration time.

03 · STATE MODEL

Which player-state transitions the combat bot exercised

The combat bot plays without god mode: it aims at visible monsters and fires, and otherwise roams. Every step is classified into a player state, and each change of state is counted against the transitions a designer would expect to see.



Transition coverage

12 of 16 expected transitions exercised across 24 episodes on 6 maps.

Not yet seen: `Hurt → Pickup`
Not yet seen: `Combat → Pickup`
Not yet seen: `Hurt → Dead`
Not yet seen: `Explore → LevelExit`

Unexpected transitions

Spawn → Combat (8), Explore → Dead (2) were observed but are not in the designer graph. Each one is either a missing edge in the spec or a bug.

Episode outcomes

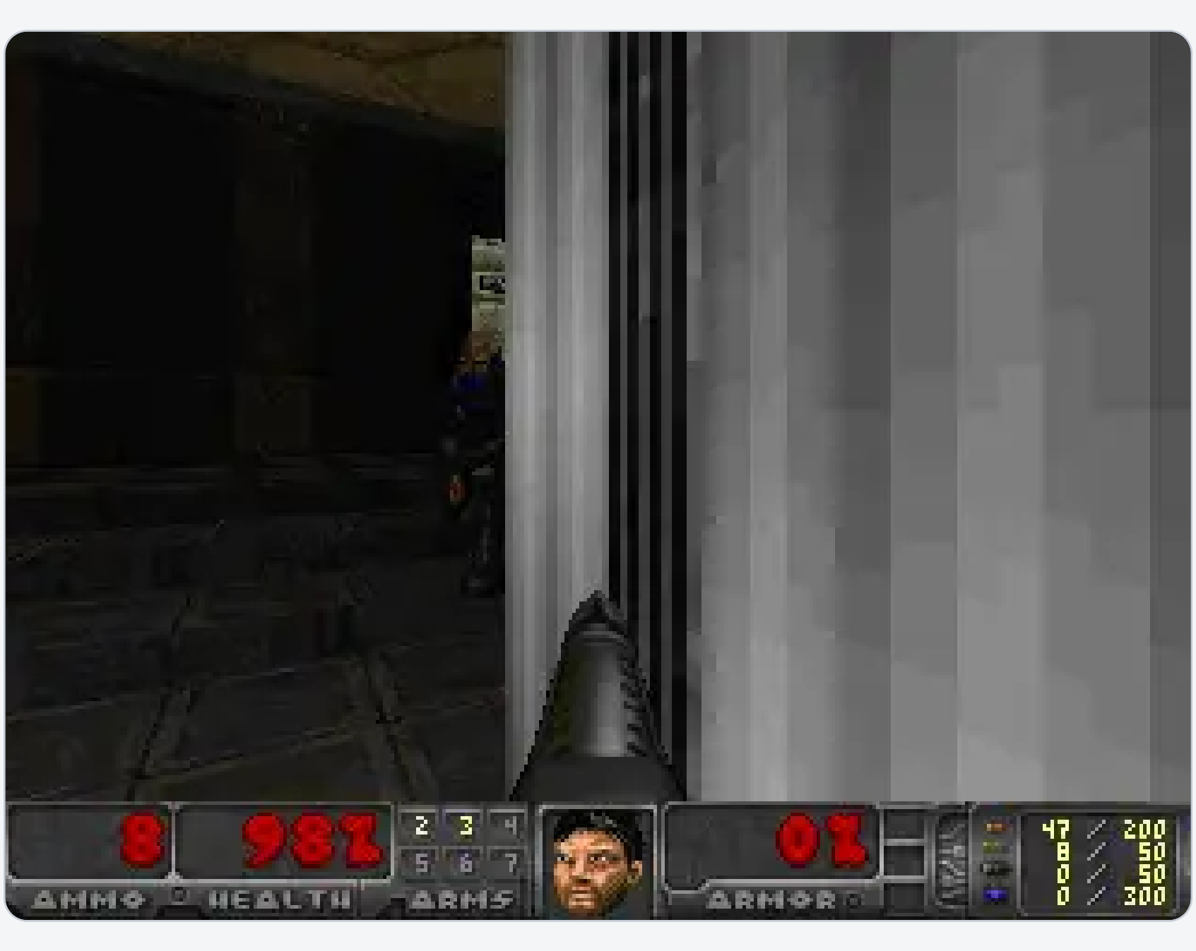
5 deaths, 19 reached the 60-second limit, 0 level exits. Kills per episode ranged 0–11. The bot is deliberately simple; a stronger persona would cover the exit and pickup edges.

04 · SMOKE MATRIX

All 32 maps boot, spawn, move, fire and hold invariants

The scripted bot runs each map for about 20 seconds of game time. Throughput was measured afterwards, one map at a time. The first, parallel run had flagged MAP28 as five times slower than the median; measured on its own it sits in the normal range, so there is no performance ticket.

Map	Checks: spawn · move · fire · use · invariants · no crash	Boot	Throughput (tics/s, serial)	Lines	Monsters	Pickups	Keys
MAP01	●●●●●● Pass	0.24 s	3,986	2,036	18	53	None
MAP02	●●●●●● Pass	0.27 s	3,684	1,717	56	86	blue card, red card, yellow card
MAP03	●●●●●● Pass	0.24 s	3,775	4,088	73	62	blue card, red card, yellow card
MAP04	●●●●●● Pass	0.22 s	4,486	2,070	123	136	blue card, red card
MAP05	●●●●●● Pass	0.26 s	4,047	2,045	76	103	blue card, red card, yellow card
MAP06	●●●●●● Pass	0.19 s	4,701	1,151	73	56	blue card, yellow card
MAP07	●●●●●● Pass	0.30 s	4,436	1,866	96	84	blue card, red card
MAP08	●●●●●● Pass	0.27 s	4,295	3,541	88	84	blue card, red card, yellow card
MAP09	●●●●●● Pass	0.29 s	3,992	3,442	140	177	blue card, red card, yellow card
MAP10	●●●●●● Pass	0.31 s	3,594	4,785	113	141	blue card, red card, yellow card
MAP11	●●●●●● Pass	0.37 s	3,043	6,533	267	374	blue card, blue skull, red card, red skull, yellow
MAP12	●●●●●● Pass	0.41 s	3,186	9,955	137	278	blue card, red card, yellow card
MAP13	●●●●●● Pass	0.23 s	4,014	2,169	115	86	blue card, red card, yellow card
MAP14	●●●●●● Pass	0.30 s	2,752	1,897	99	80	blue card, red card, yellow card
MAP15	●●●●●● Pass	0.38 s	3,234	7,472	153	175	blue card, red card, yellow card
MAP16	●●●●●● Pass	0.28 s	4,154	2,526	78	46	blue card, yellow card
MAP17	●●●●●● Pass	0.22 s	3,182	1,654	51	111	blue skull, yellow skull
MAP18	●●●●●● Pass	0.23 s	3,564	3,014	118	144	blue card, red card
MAP19	●●●●●● Pass	0.33 s	4,014	3,205	146	79	blue card, red card, yellow card
MAP20	●●●●●● Pass	0.33 s	4,222	4,794	141	130	blue card, red card, yellow card
MAP21	●●●●●● Pass	0.23 s	4,566	1,958	81	116	yellow skull
MAP22	●●●●●● Pass	0.34 s	3,921	3,761	85	104	blue card, red card, yellow card
MAP23	●●●●●● Pass	0.30 s	3,015	4,806	219	166	red skull, yellow card
MAP24	●●●●●● Pass	0.22 s	4,750	2,435	98	77	blue card, red skull, yellow skull
MAP25	●●●●●● Pass	0.42 s	3,526	7,684	122	196	blue skull, red skull, yellow skull
MAP26	●●●●●● Pass	0.31 s	3,106	3,169	242	265	blue skull, yellow skull
MAP27	●●●●●● Pass	0.32 s	4,576	5,602	140	169	blue skull, red skull, yellow skull
MAP28	●●●●●● Pass	0.43 s	2,957	10,400	194	206	blue skull, red skull, yellow skull
MAP29	●●●●●● Pass	0.31 s	3,052	1,630	97	179	blue skull, red skull, yellow skull
MAP30	●●●●●● Pass	0.24 s	3,457	464	10	78	None
MAP31	●●●●●● Pass	0.18 s	5,011	937	17	11	red card
MAP32	●●●●●● Pass	0.20 s	3,336	1,152	81	96	blue skull, red skull, yellow skull



05 · REPLAY

Every combat run is recorded as an engine demo

This clip is not a screen recording of the original run. It is 21 seconds of MAP03 episode 1 re-simulated from the demo file the combat bot wrote while playing: the engine replays the recorded inputs and produces the same game. All 24 combat episodes replayed this way and 24 matched the live run's kills, items and damage exactly. That is the repro a developer gets attached to a ticket.

06 · WHAT CARRIES OVER

What would transfer to a larger shooter

Per-commit tier

Scripted smoke bot

Boot, spawn, move, fire, use and invariant checks on all 32 maps in about a second each. Cheap enough to gate every build.

Exploration

Go-Explore with novelty weighting

Returns to rarely visited cells through save states. Over 3 seeds it beat random walk on 3 of 6 maps at equal budget, lost on 2, and tied on 1. Maps gated by keys or switches need a goal-seeking agent instead.

State coverage

Transitions against a designer graph

Player states are classified each step and counted against the transitions a designer expects, so gaps (no level exit, few pickups) show up as missing edges.

Invariants

Checks on every step

Ammo, health and armor ranges, monotonic kill and item counts, and inside-the-map and above-the-floor geometry checks.

Triage

Re-run before filing

Each candidate is replayed from its save state in a fresh process, and stuck spots get an escape test and a nearby-monster check. 9 of 12 candidates were closed: 8 test artifacts and one bug in our own checks.

Other engines

What changes on a bigger game

Replace the ViZDoom wrapper with an agent API into the game build: state read-out, actions, save/restore and replay capture (on Unreal, Gauntlet sessions plus network emulation). The checks, agents and triage flow keep their shape.

Freedoom (freedoom.github.io) and ViZDoom (vizdoom.farama.org) are open source. Exploration follows Go-Explore (Ecoffet et al., Nature 2021) and EA SEED's count-based curiosity testing (Gordillo et al., CoG 2021).

Coverage counts 64-unit map cells whose area is at least half inside a sector; some of those cells are decorative or deliberately unreachable, so 100% is not attainable. Exploration runs use god mode so agents aren't killed; combat runs do not.

Harness code, raw results, demo recordings and repro save files are in the repository alongside this page.